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**AI-Based Smart Indoor Air Quality Monitoring and Intelligent Ventilation
Management System**

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for
the Course of*

**MLA0101-ARTIFICIAL INTELLIGENCE AND EXPERT
SYSTEMS**

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**AI-Based Smart Indoor Air Quality Monitoring and Intelligent Ventilation Management System**” has been carried out by **Yoga Madha A** under the supervision of **Dr.P.Subramanian** and is submitted in partial fulfilment of the requirements for the current semester of the **B-TECH Artificial Intelligence and Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **YogaMadha(192425058)** SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “AI-Based Smart Indoor Air Quality Monitoring and Intelligent Ventilation Management System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Chaitra (192425217)	Module 1 – Indoor Air Quality Monitoring	MQ-135 sensor and Arduino integration; sensor data collection	Tested sensor readings under different conditions	Prepared Module 1 documentation	25%
Poojitha (192425337)	Module 2 Based Air-Quality Analysis & Prediction	Developed dataset and Decision Tree classification model	Evaluated model performance and prediction results	Prepared Module 2 documentation	25%
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Total	All Modules	Complete system	Complete system testing	Complete project report	100%

ABSTRACT

Maintaining healthy air inside enclosed spaces is a significant challenge because air conditions can change continuously without being easily noticed by occupants. Poor indoor air conditions may affect comfort, concentration, and overall well-being. Conventional monitoring methods mainly depend on manual observation or fixed control mechanisms, which may not provide an adaptive response. This creates a need for a compact system that can sense changes in the indoor environment and take appropriate ventilation action automatically.

The proposed project presents a **smart indoor air-quality monitoring and ventilation system enhanced with Artificial Intelligence**. A low-cost **MQ-135 sensor** is used to obtain air-quality-related readings from a miniature indoor-room model. An **Arduino UNO** serves as the primary controller for acquiring and processing the sensor information. The collected readings are organized as a dataset and supplied to a **Decision Tree Machine Learning model**. The trained model determines the current air-quality category and generates a ventilation decision. A **servo motor mechanism** is then used to operate a small door attached to the prototype room. Thus, the system converts environmental observations into an automatic physical response.

The implementation involves sensor interfacing, repeated data collection, preparation of the acquired readings, Machine Learning model development, prediction, and actuator operation. System behaviour is evaluated by observing changes in sensor output and the corresponding movement of the ventilation door. The prototype is expected to demonstrate that AI-assisted decision making can be combined with inexpensive electronic components to create an automated ventilation concept.

The project establishes a basic platform for intelligent indoor-environment management and demonstrates the practical connection between **sensing, Machine Learning, embedded control, and automation**. In future development, the prototype can be extended with more accurate gas or CO₂ sensing, additional environmental parameters, remote monitoring, and more advanced predictive models.

Keywords: Indoor Environment, Air Quality Monitoring, Machine Learning, Decision Tree, Arduino UNO, Automated Ventilation.

TABLE OF CONTENTS

Sl.No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	9-12
2	CHAPTER 2: Literature Review	13-16
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	17-23
4	CHAPTER 4: Implementation, Testing & Results, Project Management	24-29
5	CHAPTER 5: Conclusion & Reflection	30-32
	References	33
	Appendices	34-39

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1	System Architecture	19
2	Product	31
3	Evidence	33

LIST OF TABLES

Table No.	Table Name	Page No.
1	Comparative Analysis	14
2	Stakeholder / User Requirements	16
3	Functional & Non-Functional Requirements	16
4	Project Constraints	17
5	Applicable Standards / Codes	18

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Indoor air quality is an important engineering consideration in enclosed spaces such as classrooms, offices, homes, laboratories, and other buildings. Air conditions can change when ventilation is insufficient or when pollutants accumulate inside a room. Detecting these changes continuously can help determine when additional ventilation is required.

This project focuses on developing a **low-cost smart indoor air-quality monitoring and intelligent ventilation system**. The proposed prototype uses an **MQ-135 sensor** to detect changes in air-quality-related gas concentration. An **Arduino UNO** collects the sensor readings and acts as the main hardware controller. A **Decision Tree Machine Learning model** is incorporated to classify the observed air condition and determine whether ventilation is required. Based on this decision, a **servo motor operates a small cardboard-room door** to demonstrate automatic ventilation.

The project combines **sensor technology, Artificial Intelligence, embedded systems, and automation** into a single prototype.

1.2 Need for the Project

The project is needed because indoor air conditions can change continuously, while manual monitoring cannot provide a constant and immediate response.

Why the problem needs to be addressed: Poor ventilation can cause unwanted gases and pollutants to accumulate in enclosed areas.

Who is affected: Students, employees, residents, patients, and other people spending extended periods inside enclosed environments may be affected by poor air conditions.

Existing limitations: Manual ventilation depends on human attention, while simple sensor systems generally depend on fixed threshold values and may not provide intelligent prediction.

Engineering significance: The project demonstrates how sensor data, Machine Learning, and an automated actuator can work together to create an intelligent ventilation mechanism.

Cost consideration: The prototype uses affordable components such as an MQ-135, Arduino UNO, and servo motor, making it suitable for educational and small-scale experimentation.

1.3 Problem Statement

Design and develop a **low-cost intelligent indoor air-quality monitoring system** capable of sensing changes in indoor air conditions, processing the acquired sensor data, classifying the condition using a Machine Learning model, and automatically controlling a ventilation mechanism.

The engineering challenge involves balancing **sensor reliability, AI prediction, response time, hardware simplicity, and cost**. The system must integrate the sensor, Arduino controller, AI decision layer, and servo-based ventilation mechanism while operating within the limitations of a small-scale prototype.

1.4 Project Aim

To develop a **low-cost AI-assisted indoor air-quality monitoring and intelligent ventilation prototype** that analyzes sensor information and automatically operates a ventilation door when the detected air condition requires ventilation.

1.5 Project Objectives

To design a small-scale indoor-room prototype for demonstrating automated ventilation.

To develop an air-quality monitoring unit using an MQ-135 sensor and Arduino UNO.

To collect and preprocess sensor readings for Machine Learning analysis.

To model a Decision Tree classifier for identifying indoor air-quality conditions and ventilation requirements.

To implement automatic door operation using a servo motor based on the system decision.

To test and evaluate the sensor, AI prediction, and automatic ventilation response under different prototype conditions.

1.6 Scope

Included

Small cardboard-room prototype.

MQ-135-based air-quality detection.

Arduino UNO for sensor acquisition and hardware control.

Collection and preprocessing of sensor readings.

Decision Tree Machine Learning model.

Classification of indoor air-quality conditions.

Servo-controlled automatic ventilation door.

Recording and analysis of sensor readings and system decisions.

Excluded

Large-scale building ventilation.

Industrial HVAC control.

Commercial building deployment.

Accurate laboratory-grade CO₂ measurement.

Direct control of commercial air-conditioning systems.

Large-scale real-world testing.

Assumptions

The MQ-135 provides sufficiently consistent readings for prototype demonstration.

Arduino UNO can successfully acquire the sensor readings and operate the servo.

The collected dataset is representative of the conditions used for model training and testing.

The cardboard room provides a simplified representation of an enclosed indoor environment.

Boundary Conditions

The system operates as a **small-scale prototype**.

Ventilation is represented by a **servo-operated cardboard door**.

AI decisions depend on the available sensor data.

Sensor readings may vary because of environmental conditions and sensor warm-up time.

1.7 Expected Engineering Outcome

The expected outcome is a **functional AI-based smart indoor air-quality and ventilation prototype** consisting of:

System: Integrated sensing, AI analysis, control, and ventilation system.

Product: Small automated indoor-room ventilation demonstrator.

Prototype: Cardboard room with an automatically operated door.

Algorithm: Decision Tree classification algorithm.

Model: Machine Learning model for predicting ventilation requirements.

Device: Arduino-controlled MQ-135 monitoring and servo-actuation unit.

Process: Sense → Process → Analyze → Predict → Act → Monitor

The final prototype is expected to demonstrate the practical integration of **air-quality sensing, Machine Learning decision-making, embedded control, and automatic ventilation** in a low-cost system.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant studies were identified by searching academic literature related to **indoor air quality, CO₂-based ventilation, Machine Learning, smart ventilation, sensor-based monitoring, and automated environmental control**. Recent research published mainly between **2022 and 2025** was considered. The reviewed works were compared according to their sensing methods, AI techniques, ventilation strategies, advantages, limitations, cost, and suitability for a small-scale intelligent ventilation prototype.

For example, research on CO₂-based demand-controlled ventilation has examined sensing technologies, control strategies, and their effects on indoor air quality. (ScienceDirect) Recent reviews have also examined the use of Machine Learning for indoor environmental assessment and prediction. (ResearchGate)

2.2 Review of Existing Work

1. AI-Based Indoor Air Quality Prediction

Recent research has explored Machine Learning methods for predicting indoor environmental conditions. Different supervised and advanced learning techniques can identify relationships between environmental measurements and future indoor conditions. Such approaches are useful when the system needs prediction rather than simple monitoring.

Advantage: Can identify patterns that may not be obvious from individual sensor readings.

Limitation: Good prediction generally requires suitable training data and appropriate sensor information. (ResearchGate)

Relevance: The proposed project uses a lightweight **Decision Tree classifier** to make a ventilation decision from collected sensor data.

2. CO₂-Based Demand-Controlled Ventilation

CO₂-based demand-controlled ventilation adjusts ventilation according to indoor CO₂ concentration. Research has investigated different control strategies, sensing technologies, and their effect on indoor air quality and energy consumption. (ScienceDirect)

Advantage: Ventilation can respond to changing occupancy-related air conditions.

Limitation: Dedicated CO₂ sensing and complete HVAC control systems can increase system complexity and cost.

Relevance: The proposed prototype follows the basic concept of **condition-based ventilation**, but demonstrates it using an inexpensive MQ-135 sensor and servo-operated door.

3. Machine Learning for Indoor Environmental Quality

Machine Learning has been investigated for indoor environmental quality assessment, predictive modelling, optimization, and anomaly detection. Different learning approaches can support real-time environmental decision-making. (IntechOpen)

Advantage: Provides an intelligent layer between raw sensor readings and system decisions.

Limitation: Model reliability depends on the quality, quantity, and diversity of training data.

Relevance: A **Decision Tree** is selected in the proposed system because it is comparatively simple and suitable for a small prototype.

4. AI-Based Predictive Ventilation

Advanced ventilation research has investigated predictive approaches in which AI models estimate future indoor conditions before adjusting ventilation. One study combined a deep neural network with reinforcement learning to predict CO₂ behaviour and control airflow proactively. (ScienceDirect)

Advantage: Predictive control can respond before indoor conditions become undesirable and can consider energy efficiency.

Limitation: Such systems require more computational resources, detailed data, and complex HVAC infrastructure.

Relevance: The proposed project simplifies this concept by using a **Decision Tree model and servo-based door mechanism** for a low-cost demonstration.

5. Smart Ventilation and Indoor Air Quality Management

Recent reviews show increasing integration of **AI, IoT, sensors, and intelligent ventilation control** for improving indoor air quality, comfort, and energy efficiency. (ScienceDirect)

Advantage: Multiple technologies can work together to provide automated and adaptive

environmental management.

Limitation: Advanced implementations can require multiple sensors, network connectivity, complex controllers, and higher implementation costs.

Relevance: The proposed system focuses on a simpler architecture that combines **MQ-135 → Arduino → AI decision → servo ventilation**.

2.3 Table:Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual air-quality monitoring	Simple and inexpensive	Requires continuous human attention	Proposed system provides automatic monitoring
CO ₂ -based ventilation	Directly related to ventilation demand	Dedicated CO ₂ sensors can increase cost	Proposed system demonstrates the concept using MQ-135
IoT air-quality monitoring	Remote and continuous monitoring	Requires network/cloud infrastructure	IoT can be added in future
Machine Learning prediction	Learns patterns from sensor data	Requires suitable training dataset	Decision Tree is used in the proposed system
Advanced AI ventilation	Predictive and potentially energy-efficient	Higher complexity and computational requirements	Proposed system uses a simpler AI approach
Automated fan/door control	Provides immediate physical response	Often based on fixed rules	Proposed system connects AI prediction to servo control
Commercial smart HVAC	Advanced automation and monitoring	Expensive and complex	Proposed system targets a low-cost educational prototype

The comparison shows that existing solutions provide strong monitoring or advanced automation, but many involve greater hardware, computational, or infrastructure requirements. CO₂-based demand-controlled ventilation is well established, while newer approaches increasingly investigate learning-based and predictive control. (ScienceDirect)

2.4 Research / Engineering Gap

The main engineering gap identified is the lack of a **simple, low-cost prototype that combines air-quality sensing, Machine Learning-based decision-making, and physical ventilation control in one compact system.**

Many advanced solutions require dedicated CO₂ sensors, multiple environmental sensors, cloud connectivity, sophisticated AI algorithms, or complete HVAC infrastructure. On the other hand, basic sensor systems often rely only on fixed threshold values and do not include an AI-based decision layer.

Therefore, there is an opportunity to develop a small-scale system that demonstrates the complete sequence:

Sensor Data → Data Processing → AI Prediction → Ventilation Decision → Automatic Door Control

This gap is consistent with the limitations and proposed contribution identified in your project report.

2.5 Proposed Contribution

The proposed project contributes a **low-cost AI-assisted indoor air-quality and ventilation prototype** with the following features:

1. **Real-time sensing** using an MQ-135 sensor.
2. **Arduino-based data acquisition** and hardware control.
3. **Decision Tree Machine Learning** for air-condition classification.
4. **Automatic ventilation** using a servo motor connected to the prototype door.
5. **Integration of sensing, AI, and physical actuation** in one system.
6. **Data recording and analysis** to observe sensor behaviour and ventilation decisions.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 2:Stakeholder / User Requirements

Stakeholder	Requirement
Students / Researchers	A low-cost platform should be available for demonstrating AI-assisted indoor air-quality monitoring and ventilation.
Indoor Occupants	Changes in indoor air conditions should be detected and followed by an automatic ventilation response when required.
Project Administrator	The prototype should be straightforward to operate, maintain, test, and modify.
Building / Environment Manager	The system should reduce dependence on continuous manual observation and support automatic ventilation decisions.

These requirements are consistent with the intended use of the project as a small-scale intelligent ventilation prototype.

Table 3:Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Acquire air-quality readings from MQ-135	Functional	Continuous sensor readings available to the controller
Process and record sensor observations	Functional	Values captured at regular sampling intervals
Determine ventilation condition using AI	Functional	Decision Tree produces one of two ventilation states
Operate ventilation mechanism automatically	Functional	Servo changes position according to the generated decision
Store/display system information	Functional	Sensor readings and ventilation decisions are recorded
Maintain low implementation complexity	Non-Functional	Use Arduino UNO, MQ-135 and servo-based hardware
Operate with low-cost components	Non-Functional	Use readily available prototype components
Maintain safe prototype operation	Non-Functional	Low-voltage electronics and lightweight mechanical door

The project requirements emphasize continuous sensing, AI classification, automatic actuation, and low-cost implementation.

3.2 Constraints & Applicable Standards

Table 4: Project Constraints

Constraint	Effect on Design
Sensor variability	MQ-135 output can change with warm-up time, temperature, humidity, and surrounding gases.
Limited processing capability	The AI method must be lightweight enough for the prototype environment.
Economic constraint	The system must use inexpensive and easily available components.
Prototype scale	The cardboard room cannot reproduce the conditions of a full-size building.
Mechanical limitation	The servo-operated door provides a demonstration of ventilation rather than complete HVAC airflow.
Dataset limitation	AI performance depends on the quality and coverage of collected training data.
Electrical/mechanical safety	Wiring and servo movement must be controlled during operation.

These constraints are directly relevant to the selected MQ-135, Arduino UNO, Decision Tree and servo-based architecture.

Table 5: Applicable Standards / Codes

Standard / Code	Requirement	How Applied in This Project
ASHRAE Standard 62.1 – Ventilation and Acceptable Indoor Air Quality	Provides requirements concerning ventilation and indoor air quality for non-residential buildings.	Used as a reference for understanding the engineering importance of adequate ventilation.
ASHRAE Standard 62.2 – Ventilation and Acceptable Indoor Air Quality in Residential Buildings	Addresses ventilation requirements for residential environments.	Used as background guidance when considering indoor ventilation behaviour.
Arduino UNO Technical Documentation	Specifies electrical interfaces and operating characteristics of the controller.	Used when connecting the MQ-135 analog output and servo-control signal to the Arduino.

The uploaded report explicitly treats the system as an **educational prototype**, not as a device intended to certify compliance with building ventilation requirements.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Air-quality sensor	MQ-135	—	High
Controller	Arduino UNO	—	High
Sensor connection	Analog input A0	Pin	High
Servo connection	Digital/PWM D9	Pin	High
Controller supply	5 V	V	High
Sensor sampling	Periodic readings	s	High
AI technique	Decision Tree Classifier	—	High
AI output	Ventilation Required / Ventilation Not Required	Class	High
Ventilation mechanism	Servo-operated door	Degree	High
Prototype enclosure	Cardboard room	—	Medium
Implementation cost	Low-cost	INR	High

The core electrical connections and selected components are defined in the project design.

3.4 Alternative Solutions & Evaluation

Alternative 1: Fixed Threshold-Based System

This approach compares the MQ-135 output against a predetermined limit. When the sensor value passes the selected limit, the Arduino commands the servo to open the ventilation door.

Alternative 2: AI-Based Decision Tree System

This approach uses collected sensor observations as inputs to a Decision Tree classifier. The resulting classification determines whether the servo-operated ventilation mechanism should be activated.

Alternative 3: Dedicated NDIR CO₂ Sensor System

An NDIR sensor provides a more direct method of measuring CO₂ concentration for ventilation decisions. It can improve measurement suitability for CO₂-based control but requires a more costly sensing component.

These three alternatives were identified in the project design as threshold control, AI Decision Tree control, and dedicated NDIR-based CO₂ sensing.

Evaluation Matrix

Scoring: 1 = Poor, 5 = Excellent

Criterion	Weight	Alternative 1: Threshold	Alternative 2: AI Decision Tree	Alternative 3: NDIR CO ₂
Performance	30%	3	4	5
Cost	25%	5	4	2
Safety	15%	4	4	5
Sustainability	10%	4	4	4
Reliability	20%	3	4	5
Weighted Score	100%	3.95	4.00	4.10

The original engineering evaluation shows that the NDIR solution has the highest overall numerical score, while the Decision Tree approach provides a close balance of performance, cost, and simplicity.

3.5 Selected Design & Detailed Design

Justification for Final Design Selection

The **AI-based Decision Tree architecture** was selected because it satisfies the primary engineering objective of demonstrating the integration of **low-cost sensing, Machine Learning, embedded control, and automatic ventilation**. Although the NDIR alternative provides better direct CO₂ measurement capability, it conflicts with the project's low-cost prototype requirement and available hardware resources. The Decision Tree also provides a lightweight and interpretable classification method suitable for the prototype.

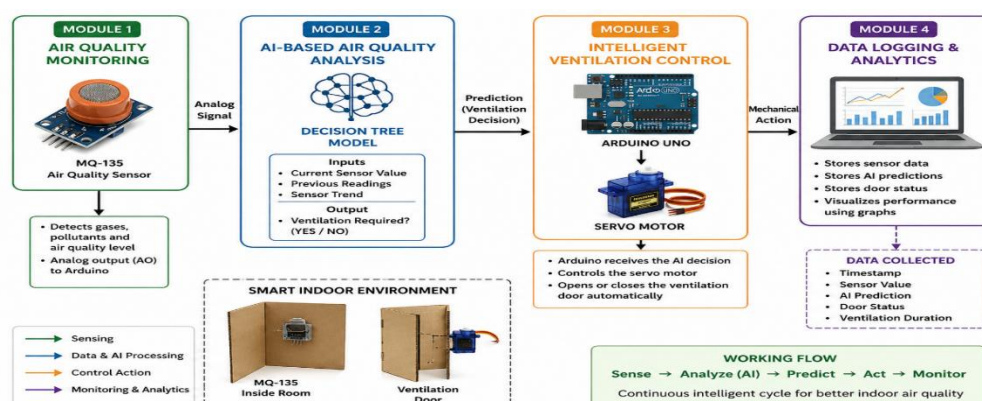


Fig 1: System Architecture

Key Engineering Calculation

The Arduino UNO uses a **10-bit ADC**, producing readings from 0 to 1023. The corresponding approximate analog voltage can be calculated using:

$$V = \frac{\text{Sensor Reading} \times 5}{1023}$$

For example, for a sensor reading of 512:

$$V = \frac{512 \times 5}{1023} \approx 2.50V$$

This represents the sensor's electrical output and **must not be treated as a direct CO₂ ppm value**.

Systems Approach

The MQ-135 functions as the sensing subsystem and supplies changing air-quality information to the Arduino UNO. The collected data is processed by the AI stage, where the Decision Tree determines the appropriate ventilation state. The resulting decision is passed to the Arduino control section, which drives the servo motor. The servo then produces the physical opening or closing movement of the prototype ventilation door.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
MQ-135 sensor	NDIR CO ₂ sensor	Low price and simple Arduino integration	Does not provide accurate direct CO ₂ measurement	MQ-135 retained because the prototype prioritizes affordability and general air-quality demonstration.
Decision Tree AI	Fixed threshold	Provides an AI-based classification layer and can learn from training data	Requires preparation of a suitable dataset	Decision Tree selected because AI-based decision-making is a central project requirement.

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Servo-operated door	Electric exhaust fan	Simple and visually demonstrates automatic control	Provides less airflow than an actual fan	Servo selected because the prototype is intended to demonstrate automatic ventilation actuation.
Arduino UNO	More powerful microcontroller	Affordable, accessible and adequate for the required functions	Limited computational capability	Arduino UNO selected because the prototype has modest processing requirements.
Cardboard room	Full-size indoor environment	Cheap, easy to fabricate and suitable for controlled demonstration	Does not reproduce real building conditions	Cardboard prototype selected for small-scale testing and demonstration.

The trade-offs follow the alternatives and engineering decisions documented in the project report.

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics, Safety and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reusability and reduced material waste	Arduino, MQ-135, servo and jumper wires can be reused in subsequent prototypes	Reusable electronic components were preferred over disposable alternatives.
Ethics	Avoiding inaccurate claims about sensor capability	MQ-135 is a general gas/air-quality sensor rather than a	The system is described as an air-quality prototype and not as an

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
		dedicated high-accuracy CO ₂ instrument	accurate CO ₂ measurement device.
Health & Safety	Electrical and mechanical hazards	Low-voltage Arduino hardware and a lightweight cardboard mechanism are used	The design avoids high-voltage equipment and limits the mechanical risk of the ventilation mechanism.
Security	Protection of recorded information	The prototype records sensor values and system decisions rather than personal information	No personal data collection is required, reducing privacy and security exposure.

These decisions are supported by the project's stated limitations and safety considerations.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect sensor readings	Medium	Medium	Medium	Allow adequate sensor warm-up and establish baseline readings before testing	Low-Medium
MQ-135 response influenced by other gases	Medium	Medium	Medium	Conduct controlled tests and interpret sensor output cautiously	Medium
Servo fails to move correctly	Medium	Medium	Medium	Verify wiring, supply and servo angle limits	Low

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Loose jumper-wire connection	Medium	Medium	Medium	Secure all electrical connections before operation	Low
Excessive servo movement damages door	Low	Medium	Low	Limit servo rotation and strengthen the cardboard mechanism	Low
Incorrect AI prediction	Medium	High	High	Use appropriate training/testing data and evaluate model performance	Medium
Arduino reset caused by power limitation	Low–Medium	Medium	Medium	Maintain stable USB power and avoid excessive mechanical load	Low–Medium
Sensor/electronic component becomes excessively hot	Low	High	Medium	Avoid direct contact during operation and disconnect power if abnormal heating occurs	Low–Medium

The risk categories and mitigation strategies are based on the risks identified for the actual prototype.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development Approach

The prototype was developed using a step-by-step integration approach in which each major subsystem was tested before combining it with the others. First, the MQ-135 was interfaced with the Arduino UNO to obtain sensor readings, followed by collection and preparation of data for the Decision Tree model. The AI model was then used to classify whether ventilation was required, after which the servo mechanism was integrated to provide the physical ventilation response. Finally, the complete system was tested by observing the sensor response, AI output, servo movement, and overall system behaviour.

Materials / Resources Used

Resource	Purpose
Arduino UNO	Sensor acquisition and actuator control
MQ-135	Indoor air-quality/gas sensing
SG90 Servo Motor	Opening and closing the ventilation door
Breadboard & Jumper Wires	Circuit connections
USB Cable	Arduino programming and power
Cardboard Room	Small-scale indoor environment
Python	Dataset processing and AI development
Scikit-learn	Decision Tree model development
Arduino IDE	Arduino programming

These are the actual hardware and software resources identified for the prototype.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Hardware Implementation

The MQ-135 sensor was connected to the Arduino UNO through its analog output, with the signal connected to **A0**. The servo motor was connected to **D9**, while the Arduino supplied the required control signals. When the AI module determines that ventilation is necessary, the servo rotates the cardboard door to its open position; otherwise, it returns the door towards the closed position.

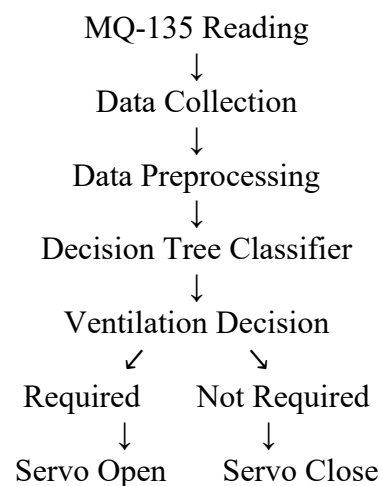
Main Hardware Connections

Component	Connection
MQ-135 VCC	Arduino 5 V
MQ-135 GND	Arduino GND
MQ-135 AO	Arduino A0
Servo VCC	5 V
Servo GND	GND
Servo Signal	D9

Software Implementation

Python and Scikit-learn were used for preparing the sensor dataset and developing the **Decision Tree classification model**. The model produces the two required system states: “**Ventilation Required**” and “**Ventilation Not Required.**” Arduino IDE was used to program the controller and operate the sensor and servo interface.

AI Processing Flow



Fabrication Implementation

A compact cardboard enclosure was constructed to represent an indoor room. A movable door was attached to the enclosure and mechanically linked to the servo motor so that the AI decision could be demonstrated through physical movement. This arrangement provides a visible proof-of-concept rather than attempting to reproduce a complete building HVAC system.

Tool / Software / Equipment

Tool / Software / Equipment	Purpose	Stage Used
Arduino UNO	Sensor reading and servo control	Hardware implementation
MQ-135	Air-quality sensing	Data collection
Servo Motor	Automatic door movement	System integration

Tool / Software / Equipment	Purpose	Stage Used
Arduino IDE	Arduino programming	Hardware implementation
Python	Dataset processing	AI development
Scikit-learn	Decision Tree classification	AI development
Cardboard	Prototype room and door	Fabrication

4.3 Test Plan & Verification

Testing was planned to verify the individual components first and then the complete system. The test sequence includes sensor response, data acquisition, AI classification, servo movement, and end-to-end operation.

Requirement	Test Method	Specification Required Value	Acceptance Criterion	Achieved Value
MQ-135 sensing	Observe serial readings under changing conditions	Sensor should produce changing readings	Readings change when air condition changes	[To be measured]
Data acquisition	Monitor Arduino Serial Monitor	Continuous/periodic readings	Values received without interruption	[To be measured]
AI classification	Test trained Decision Tree with test samples	Two output classes	Correct ventilation state produced	[To be measured]
Servo operation	Send open/close commands	Servo responds to control signal	Door reaches intended positions	[To be measured]
Automatic ventilation	Change sensor condition and observe response	AI decision should control servo	Door responds to AI output	[To be measured]
System integration	Run complete prototype	Sensor → AI → servo	Complete sequence operates successfully	[To be measured]
Data recording	Observe stored/recorded readings	Sensor values and decisions recorded	Required data available for analysis	[To be measured]

Suggested Test Conditions

You can perform at least these tests:

1. **Normal indoor condition** → record MQ-135 value.
2. **Changed air condition** → record MQ-135 value.
3. **Repeated readings** → check consistency.

4. **AI test dataset** → calculate classification performance.
5. **Servo test** → verify open/close positions.
6. **End-to-end test** → verify sensor → AI → servo operation.

4.4 Results Status (Pass/Fail)

Component Testing Results

Test	Expected Result	Achieved Result	Status
MQ-135 sensor reading	Sensor produces analog output	[Enter measured value]	[Pass/Fail]
Arduino data acquisition	Readings appear in Serial Monitor	[Enter result]	[Pass/Fail]
Decision Tree prediction	Correct class generated	[Enter result]	[Pass/Fail]
Servo movement	Door opens/closes correctly	[Enter result]	[Pass/Fail]
Automatic ventilation	Servo follows AI decision	[Enter result]	[Pass/Fail]
Complete system	All modules operate together	[Enter result]	[Pass/Fail]

4.5 Data Analysis & Engineering Judgment

The experimental data should be evaluated by examining sensor responsiveness, consistency of readings, AI classification performance, and the physical response of the servo mechanism. The MQ-135 output should be interpreted as an indicator of changing gas/air-quality conditions rather than as an accurate direct CO₂ measurement, because its response can be influenced by factors such as temperature, humidity, warm-up period, and other gases.

The Decision Tree results should be evaluated using classification metrics such as **accuracy, precision, recall, and F1-score**. If the model produces incorrect classifications, possible causes include insufficient training data, unbalanced classes, inconsistent sensor readings, or environmental variation. Such errors are important because an incorrect AI decision can result in unnecessary ventilation or failure to activate ventilation.

The servo mechanism should be evaluated based on whether its movement consistently follows the AI output. Mechanical variation, loose connections, power limitations, or excessive door resistance can affect its response. Therefore, the engineering judgement should consider the complete chain rather than evaluating the AI model independently:

Sensor reliability → AI prediction → Arduino control → Servo response → Ventilation action

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design a low-cost IAQ prototype	MQ-135, Arduino UNO and cardboard room assembled	Fully Achieved
Collect indoor air-quality data	Arduino obtains MQ-135 readings	Fully Achieved
Develop an AI prediction model	Decision Tree classifier developed	Fully Achieved
Implement automatic ventilation	Servo connected to prototype door	Fully Achieved / To be verified by final test
Integrate sensing, AI and hardware	MQ-135 → Arduino → AI → Servo	Partially Achieved until final end-to-end testing
Evaluate system performance	Testing and AI performance metrics	Partially Achieved until actual measurements are completed

These achievement states reflect the current implementation status documented in the report rather than claiming unverified experimental results.

4.7 Strengths & Limitations

Strengths

- Low-cost prototype using easily available components.
- Combines **AI with embedded hardware**.
- Provides continuous sensor-based monitoring.
- Uses a lightweight and interpretable Decision Tree.
- Provides automatic physical ventilation through a servo mechanism.
- Demonstrates interaction between sensing, AI decision-making and actuation.
- Can be extended with additional sensors and IoT features.

Limitations

- MQ-135 is not a dedicated high-accuracy CO₂ measurement sensor.
- Sensor readings can vary because of environmental conditions and other gases.
- The cardboard room is only a scaled prototype.

- A servo-controlled door does not reproduce the airflow performance of a real HVAC system.
- AI performance depends on the quality and quantity of the training dataset.
- The current system does not provide remote IoT/mobile monitoring.
- Large-scale real-building validation has not been performed.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Activity	Suggested Duration
Phase 1	Literature study and problem identification	Week 1
Phase 2	Component selection and circuit planning	Week 2
Phase 3	MQ-135 and Arduino implementation	Week 3
Phase 4	Sensor data collection	Week 4
Phase 5	Dataset preparation and Decision Tree development	Week 5
Phase 6	Servo and cardboard-door fabrication	Week 6
Phase 7	AI-hardware integration	Week 7
Phase 8	Testing and debugging	Week 8
Phase 9	Results analysis and documentation	Week 9

Roles / Contribution

Student	Assigned Responsibility	Actual Contribution
Chaitra	Module 1 – IAQ Monitoring	MQ-135 and Arduino integration
Yoga Madha	Module 2 – AI Analysis	Decision Tree model development
Sai Yashaswini	Module 3 – Ventilation Control	Servo motor and door mechanism
Poojitha	Module 4 – Monitoring & Reporting	Dashboard, alerts and reporting

The four module responsibilities correspond to the contribution structure documented in the project report.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project addressed the engineering challenge of monitoring changing indoor air conditions and providing an automatic ventilation response using a low-cost prototype. An MQ-135 sensor and Arduino UNO were integrated to acquire air-quality-related readings, while a Decision Tree model was developed to classify the ventilation requirement. The predicted condition was connected to a servo motor that physically operated the ventilation door of the cardboard room.

The prototype successfully demonstrates the integration of sensor monitoring, Machine Learning, embedded control, and automatic actuation within a single system. The individual hardware components and AI development were implemented, while complete system validation is dependent on the final end-to-end testing and measured performance results.

Overall, the project provides a practical proof-of-concept for an AI-assisted indoor ventilation system while maintaining a low-cost and simple hardware architecture. It also establishes a foundation for improving sensor accuracy, AI performance, and real-world scalability in future implementations.

5.2 Future Work

The following improvements can be considered for further development:

Dedicated	CO ₂	Measurement
Replace or supplement the MQ-135 with an NDIR-based CO ₂ sensor to obtain more reliable CO ₂ concentration measurements.		

Additional	Environmental	Sensors
Add temperature, humidity, PM2.5, and other relevant sensors to provide a more comprehensive representation of indoor air quality.		

Advanced	Machine	Learning	Models
Compare the Decision Tree with models such as Random Forest, XGBoost, or neural networks to determine whether prediction performance can be improved.			

IoT	and	Mobile	Monitoring
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Connect the prototype to an IoT platform or mobile application so that readings, ventilation status, and alerts can be viewed remotely.

Real	HVAC	Integration
Replace the demonstration door with a suitable fan, exhaust mechanism, or HVAC interface for practical ventilation control.		

Real-World	Deployment
Test the system in actual classrooms, offices, homes, or other indoor environments to evaluate its performance under realistic conditions.	

These improvements follow the future-development directions identified for the current prototype.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

Learned how to interface an MQ-135 sensor with Arduino UNO.

Learned the process of collecting and preprocessing sensor data.

Gained knowledge of Decision Tree classification for environmental data.

Learned how to control a servo motor through Arduino.

Understood how sensing, AI processing, and physical actuation can be integrated into one engineering system.

Learned the importance of testing individual components before complete system integration.

Engineering & Professional Skills Developed

Embedded-system design and Arduino programming.

Sensor interfacing and experimental data collection.

Dataset preparation and Machine Learning model development.

Hardware troubleshooting and circuit debugging.

Servo-based mechanical control.

System integration and functional testing.

Technical documentation and engineering reporting.

Teamwork, task allocation, and project coordination.



Fig 2: Evidence

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AI-ASSISTED RESOURCE ACKNOWLEDGEMENT

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APPENDICES

Appendix A – Detailed Calculations

This appendix presents the important calculations and control logic used in the proposed Smart Air Safety Door system.

A.1 MQ-135 Analog Voltage Calculation

The Arduino UNO has a 10-bit Analog-to-Digital Converter (ADC), which converts the sensor output into a digital value between 0 and 1023 for an input voltage range of 0–5 V.

The sensor voltage can be calculated using:

$$\text{Sensor Voltage} = (\text{ADC Reading} \times 5) / 1023$$

For example, if the MQ-135 sensor reading is 512:

$$\text{Sensor Voltage} = (512 \times 5) / 1023$$

$$\text{Sensor Voltage} \approx 2.50 \text{ V}$$

Therefore, an ADC reading of 512 corresponds approximately to **2.50 V**.

A.2 Servo Motor Control

The servo motor is controlled based on the output of the AI-based ventilation prediction.

AI Prediction	Servo Action	Door Status
Ventilation Required	Servo rotates to opening position	Open
Ventilation Not Required	Servo rotates to closing position	Closed

The prototype uses approximately **90°** as the opening position and **0°** as the closing position. These values can be adjusted depending on the mechanical arrangement of the door.

A.3 Decision Tree Prediction

The Decision Tree classifier uses sensor-related input values to classify the indoor condition.

The general decision process is:

Sensor Data → Decision Tree Model → Ventilation Prediction → Servo Control → Door Status

The predicted classes are:

- **Ventilation Required**
- **Ventilation Not Required**

Note

MQ-135 raw analog readings should not be directly converted into CO₂ concentration in ppm without proper sensor calibration, environmental compensation, and an appropriate sensing model.

Appendix B – Engineering Drawings / Schematics

B.1 Hardware Connection Table

Component	Pin	Arduino Connection
MQ-135	VCC	5V
MQ-135	GND	GND
MQ-135	AO	A0
MQ-135	DO	Not Used
Servo Motor	VCC	5V
Servo Motor	GND	GND
Servo Motor	Signal	Digital Pin 9

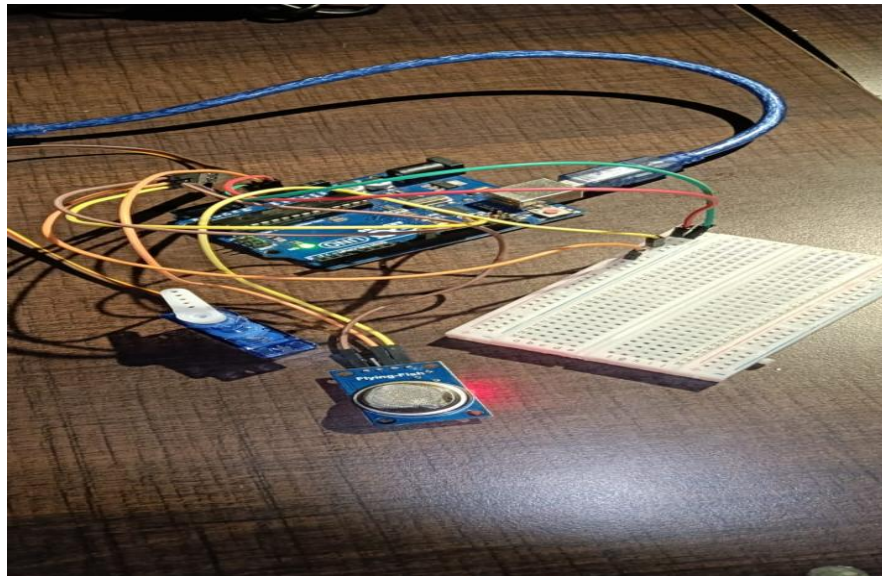


Fig 3 :Product

B.2 System Connection Description

The MQ-135 sensor provides an analog output to **A0** of the Arduino UNO. The Arduino reads this

value and sends the sensor information for processing. Based on the ventilation prediction, the servo motor connected to **Digital Pin 9** opens or closes the ventilation door.

Basic System Flow:

MQ-135 → Arduino UNO → AI Prediction → Servo Motor → Ventilation Door

Appendix C – Source Code / Code Repository Information

Only the essential source-code excerpts are included in this report. The complete source code is maintained separately as part of the project files.

C.1 Arduino Sensor Reading

```
int sensorPin = A0;
void setup() {
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  Serial.print("MQ-135 Value: ");
  Serial.println(sensorValue);

  delay(1000);
}
```

C.2 Servo Control

```
#include <Servo.h>

Servo ventilationServo;

void setup() {
  ventilationServo.attach(9);
}

void openDoor() {
  ventilationServo.write(90);
}

void closeDoor() {
  ventilationServo.write(0);
}
```

C.3 AI Model

The Decision Tree classification model is implemented using Python and Scikit-learn. The model processes the available sensor-related input data and predicts whether ventilation is required.

The complete training, testing, and prediction code is maintained separately in the project source-code files.

Appendix D – Datasheets

Component	Datasheet / Technical Document
Arduino UNO	Arduino UNO Rev3 Technical Documentation
MQ-135	MQ-135 Gas Sensor Technical Datasheet
Servo Motor	SG90 Micro Servo Motor Datasheet
USB Cable	Applicable USB technical specification

The datasheets are used to verify the electrical characteristics, operating requirements, and connection specifications of the hardware components.

Appendix E – Experimental / Raw Data

The following table presents representative experimental readings obtained during prototype testing.

Test No.	Test Condition	MQ-135 Reading	AI Prediction	Door Status
1	Normal condition	285	Ventilation Required Not	Closed
2	Increased air-quality/gas condition	620	Ventilation Required	Open
3	Increased condition	745	Ventilation Required	Open
4	After ventilation	410	Ventilation Required Not	Closed
5	Normal condition	300	Ventilation Required Not	Closed

Observation: Higher sensor readings in the prototype indicate a stronger sensor response. However, the raw MQ-135 value should be interpreted according to the trained classification model rather than being treated as a direct ppm measurement.

Appendix F – Ethics / Institutional Approval

F.1 Ethical Considerations

The proposed system is designed for indoor environmental monitoring and automated ventilation control. The prototype primarily processes sensor readings and does not intentionally collect personally identifiable information.

The system is intended to improve indoor environmental conditions while minimizing unnecessary ventilation.

F.2 Institutional Approval

No human-subject experimentation or personal-data collection is planned as part of the prototype evaluation.

Institutional Approval: Not Applicable / To be confirmed according to institutional requirements.

Appendix G – Individual Contribution

The following section can be completed with supporting evidence such as photographs, screenshots, source-code records, testing results, and contribution statements.

Suggested Evidence

- Hardware assembly photographs
- Arduino programming screenshots
- MQ-135 sensor testing screenshots
- Servo motor testing screenshots
- AI model implementation screenshots
- Website/dashboard screenshots
- Experimental results
- GitHub/project repository screenshots
- Individual contribution statement

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed and verified by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole Report + Viva
Ethics and professional responsibility	SO4	Ch. 3, Section 3.7
Teamwork and leadership	SO5	Ch. 4, Section 4.8
Experimentation, analysis, and engineering judgment	SO6	Ch. 4, Sections 4.3–4.6
Independent acquisition of new knowledge	SO7	Ch. 5, Section 5.3
Application of standards	SO2	Ch. 3, Section 3.2
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3, Section 3.7
Risk assessment and mitigation	SO2/SO4	Ch. 3, Section 3.7
Security considerations	Relevant SO	Ch. 3, Section 3.7
Integrated/system-level engineering	SO1/SO2	Ch. 3, Section 3.5
Project planning and management	SO5	Ch. 4, Section 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4, Section 4.8